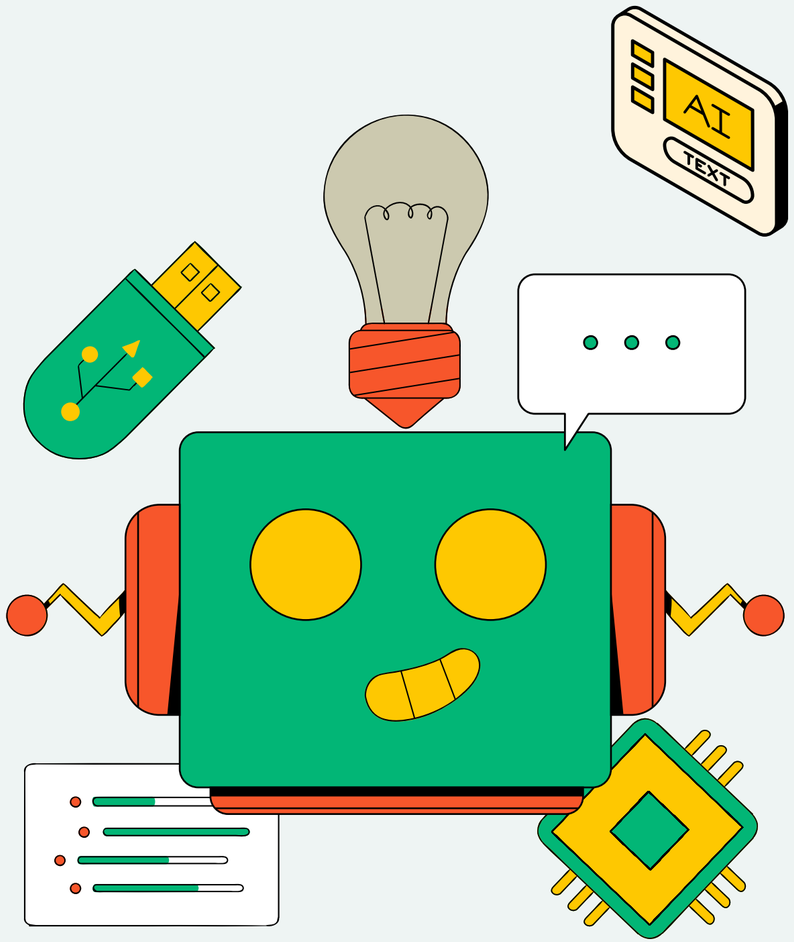
DS 301 Midterm Project – Cornerstone College

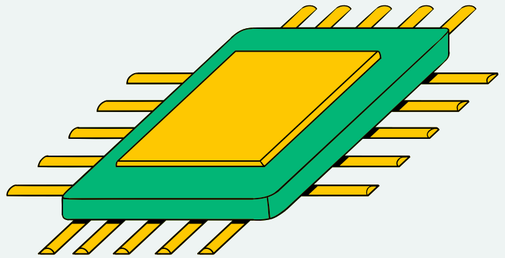


ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

PRESENTATION

PRESENTED BY:

BALBINO RAMOS



PROJECT OVERVIEW

- Objective

Develop a predictive model to classify individuals as having annual income above or below $50,000 based on demographic and socioeconomic factors.

- Problem Type

This is a binary classification problem using supervised learning techniques.

- Dataset Source

We utilize a publicly available census-based dataset from the UCI Machine Learning Repository.

FOUNDATIONAL RESEARCH

Research Paper

Supervised Learning for Binary Classification on US Adult Income

Author

Li-Pang Chen

Year

2021

This paper provided the methodological foundation for our project, evaluating multiple classification approaches including Logistic Regression and Support Vector Machines. The study demonstrated practical applications of supervised learning for income prediction using census data.

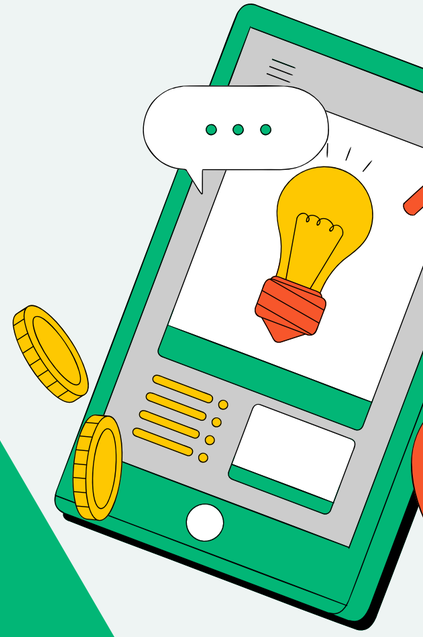
DATASET DETAILS

Adult / Census Income Dataset
Source: UCI Machine Learning Repository

48,842	14	1
Records	Predictors	Target
Total instances in dataset	Feature variables used	Binary income classification

The dataset contains a mix of **categorical and integer data types**, representing demographic information such as age, education, occupation, and marital status.

DATA PROCESSING PIPELINE



01
data loading

02
initial inspection

03
missing data handling

04
feature encoding

05
feature scaling

06
data splitting

EXPLORATORY DATA ANALYSIS

SUMMARY STATISTICS

Generated descriptive statistics

CLASS DISTRIBUTION

Examined balance between income classes

VISUAL ANALYSIS

Visualize distributions of key demographic variables

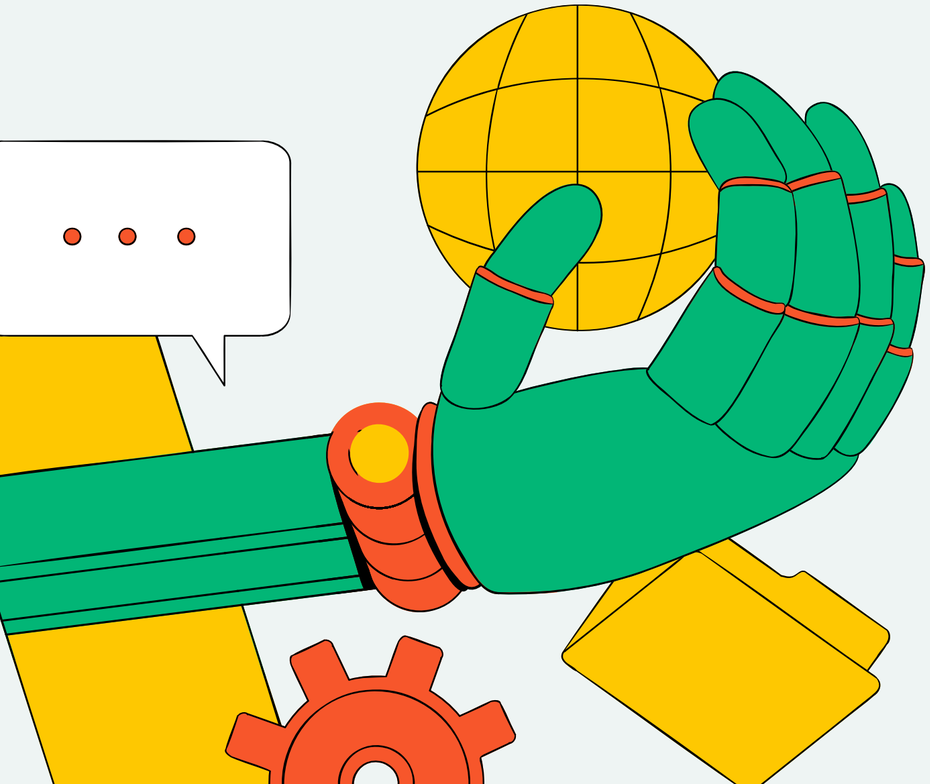
CORRELATION MATRIX

Built correlation matrix

OUTLIER DETECTION

Applied IQR rule

METHODOLOGY



Model Selection

We implemented two established classification algorithms:

Logistic Regression

Linear model for binary classification with interpretable coefficients

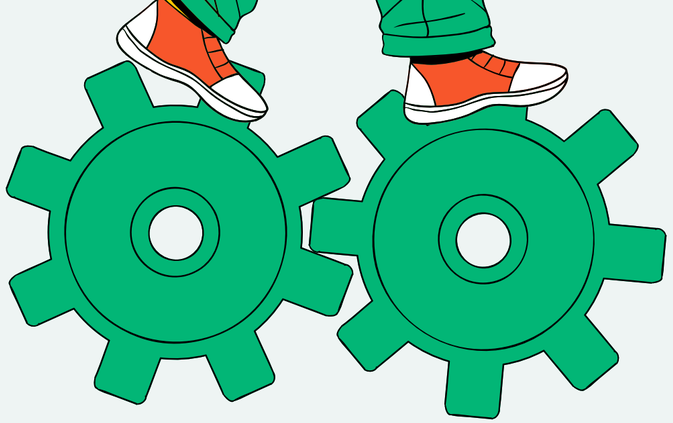
Support Vector Machine

Powerful kernel-based method for finding optimal decision boundaries

Implementation Approach

- Built scikit-learn pipelines for reproducible preprocessing
- Applied k-fold cross-validation to assess model stability
- Used GridSearchCV for systematic hyperparameter tuning of SVM (parameter C)
- Ensured consistent preprocessing across both models

MODEL IMPLEMENTATION



PREPROCESSING TIMELINE

Constructed unified pipeline incorporating imputation, one-hot encoding, and feature scaling

DATA SPLITTING

Divided dataset into training (80%) and testing (20%) sets with stratification

LOGISTIC REGRESSION

Trained baseline model with default parameters using preprocessed training data

SVM TRAINING

Trained SVM with GridSearchCV for hyperparameter optimization on training set

TEST PREDICTIONS

Generated predictions on test set for both models to evaluate real-world performance

RESULTS COMPARISSON

Both models demonstrated strong performance on the test dataset, with comparable accuracy metrics. Detailed analysis revealed subtle differences in precision, recall, and F1-score across income classes.

Test Accuracy

Both models achieved high accuracy on unseen test data, with minimal overfitting observed

Classification Reports

Generated comprehensive reports showing precision, recall, and F1-score for each income class

Cross-Validation

Evaluated cross-validation accuracy to assess model robustness and stability

The **results table** provided clear comparison of all metrics side-by-side for final model selection.

IMPROVEMENTS

01

Add feature selection to reduce dimensionality

02

Tune more hyperparameters for both models

03

Test KNN or Decision Tree for additional comparison

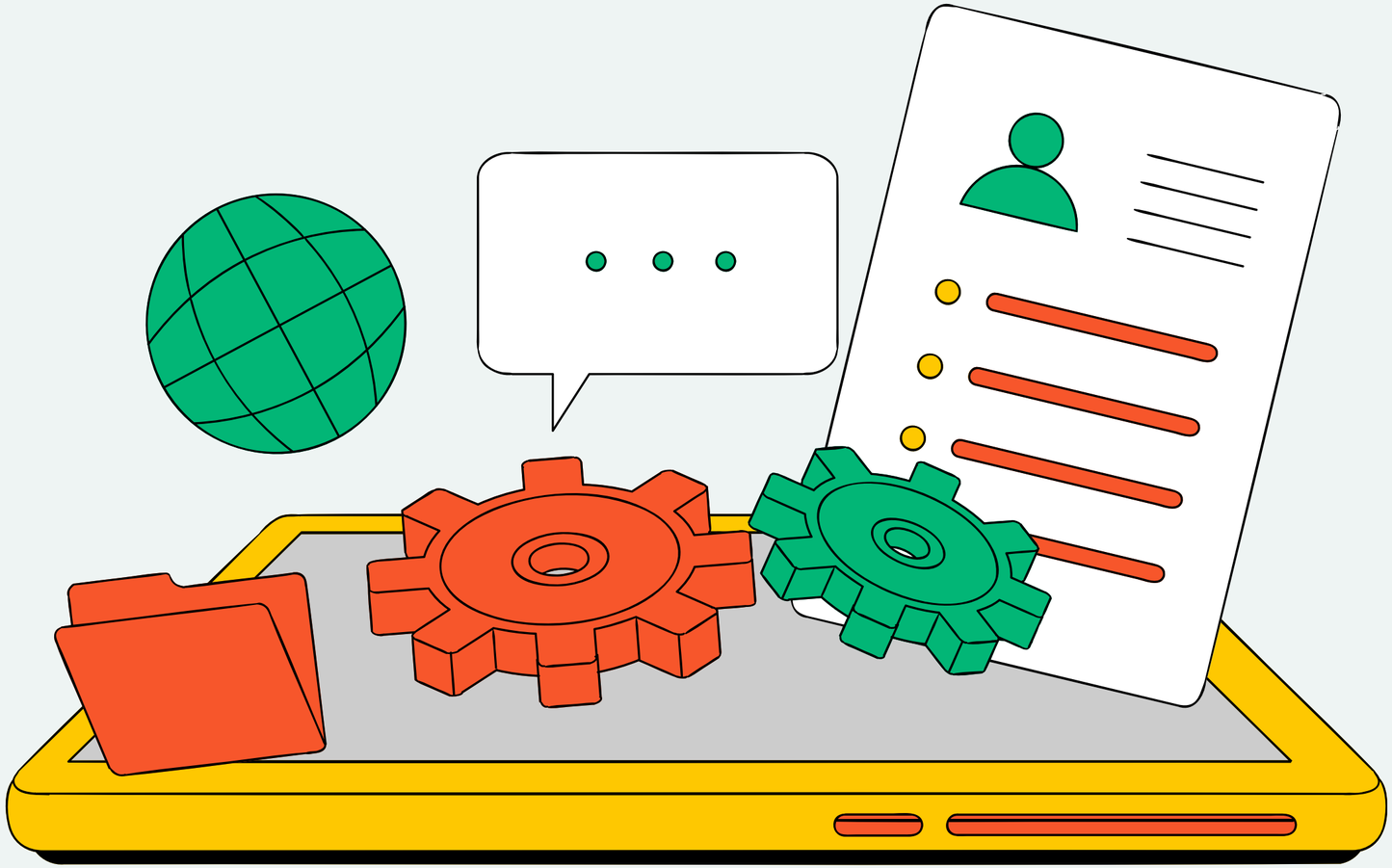
04

Address class imbalance more explicitly

05

Try stronger categorical encoding strategies

CONCLUSIONS

- Completed a full classification workflow
- Used a real paper and a real public dataset
- Implemented two approved machine learning models
- Produced a reproducible notebook-based submission

THANK YOU